

Wm. Keith van der Meulen

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Data engineer with an extensive background working with petabyte-scale datasets and analytics-heavy workloads. Experienced at designing and optimizing for highly efficient pipelines with standardized, reusable, and understandable code structure. Skilled at designing high quality data models for use in AI, analytics, data science, and machine learning.

Education

University of Calgary

Sep 2012 – Apr 2018
Computer Science, B.Sc.
Electrical Engineering, B.Sc.
Computer Engineering, minor

Skills

Technical Expertise Areas

Data pipelining, ETL workloads, dataset curation, medallion architectures, query optimization, database design, cloud architecture, data privacy, data governance

Programming Languages

Python, SQL, JavaScript, HTML, CSS, C, C++, C#, Java

Cloud Services

Databricks, Microsoft Azure, Google Cloud Platform, Amazon Web Services

Databases

Snowflake, Delta Lake, SQL Server, Google Big Query

Frameworks

Apache Spark/PySpark, GraphQL, CloudEvents

Software Products

Fivetran, Jira, PyCharm, Git

Technical Interests

Data science, computational linguistics, online communities, home automation, open-source knowledge platforms, economics, mathematics, computer vision, robotics, synthetic biology, rocketry

Work Experience

Senior Data Engineer

Stack Overflow

Jan 2022 – Present

- Lead data ingestion strategy and prioritization of first party data sources for use in generative AI, search algorithms, and click-based advertising
- Author internal company-wide Kafka-compliant event messaging specification and framework for use in AI products, data streaming, and microservices
- Build automated jobs for curating and delivering datasets for AI model training to customers, including OpenAI and Google, in data licensing deals
- Automate usage of Azure Data Lake metadata to create and maintain Snowflake warehouse schemas, data, and roles for company-wide data access
- Develop hierarchical role-based access patterns for flexible data governance
- Design cloud-based data architecture to deprecate on-premise data processing servers, event collection systems, and A/B user testing tools

Data Engineering

Accenture AI

Aug 2019 – Jan 2022

Data Engineer

Meta (Facebook)

May 2021 – Jan 2022

- Build financial forecasting pipelines in SQL for data science & analytics teams
- Explore datasets for discovering new predictive insights across terabyte-scale datasets and productionize results
- Refactor financial tables to improve data quality for data science teams

Lead Analytics Engineer

Accenture Research

Aug 2020 – Apr 2021

- Manage two separate projects simultaneously with data engineers, analysts, and QA testers spread across multiple continents
- Coordinate business managers' objectives into technical requirements
- Optimize Spark jobs to reduce run time by up to 95% on customer datasets
- Unify multiple codebases into one cohesive, standardized internal library
- Refactor Python codebase to zero linting and static code analysis errors

Data Engineer

Jones Lang LaSalle, Inc. (JLL)

Oct 2019 – Jul 2020

- Produce automated data ingestion tools on Azure Functions for 3rd party data
- Develop enterprise GraphQL API that generates itself from an external data schema configuration file using Python meta-programming techniques
- Write job orchestration scripts for running daily Databricks jobs

Data Engineer

Talroo, Inc.

Feb 2019 – Apr 2019

- Build production ETL notebooks for generating datasets from raw data streams
- Explore terabyte scale datasets with Spark for geographic trends for targeted advertising opportunities across multiple market segments
- Construct Docker-based React/Flask/SQL web app for creating labelled datasets

ML Engineer

Hotchkiss Brain Institute

May 2018 – Aug 2018

- Train and test machine learning (ML) networks to analyze MRIs for brain lesions
- Create Python-based ML models with TensorFlow and Keras frameworks
- Analyze large medical datasets to find statistical correlations and prep for ML